Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Polyurethane dry-lamination adhesive for Li-ion battery pouch films			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical / Electronics		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Fine Chemicals	Adhesives / Sealants
Concept of Proposed Technology	**Concept and Background** Lithium-ion pouch cells rely on aluminum-laminated multi-layer films such as polyethylene terephthalate/polyamide/aluminum/adhesive-stacks (PET/PA/Al/adhesive-stacks) to provide moisture/gas barrier performance, mechanical robustness, and electrical insulation [1,2,4]. During pouch forming (deep-drawing / press forming), the laminate is subjected to severe bending and biaxial stretching, especially at punch corners and radii. If interlayer adhesion is insufficient or the adhesive layer is too brittle, failures such as interfacial delamination and cohesive micro-cracking can occur, reducing yield and potentially degrading barrier integrity [3,4].			

	Industrial trends toward higher cell capacity increase the need for deeper forming while maintaining robust adhesion throughout the forming process. Recent formability studies of PET/PA/Al-based pouch films also highlight that stronger, more damage-tolerant interfaces delay strain localization and improve formability [4]. **Core technical gap (problem statement)** Conventional laminating adhesives can meet basic peel requirements for Al-to-PA bonding but may not withstand the combined demands of (i) deep forming (>6–7 mm), (ii) elevated-temperature peel performance, and (iii) durability under humidity/electrolyte exposure. The central challenge is to achieve high peel strength and heat resistance without increasing crosslink density/modulus to the point that the adhesive layer cracks during forming [4,6,7]. **Polyurethane (PU) laminating adhesives for PET/PA/Al** Two-component (2K) reactive PU adhesives are widely used in dry lamination for flexible multi-material structures because they provide strong, durable bonding and can be processed on roll-to-roll coating/lamination lines [5]. PU chemistry is highly tunable: by controlling polyol type/molecular weight, hard/soft segment ratio, isocyanate functionality, chain extenders, and catalysts, the cured network's modulus, elongation, toughness, and heat resistance can be systematically designed [6,7]. Aluminum foil surfaces present oxide/hydroxyl functionalities, while PA and PET contain polar amide and ester groups. Urethane/urea groups enable strong polar interactions (e.g., hydrogen bonding) and good wetting on these substrates, making PU a practical platform to achieve high peel strength across Al & PA and PA & PET interfaces without relying on thick tie-layers [6,7]. Battery pouch film patents and industrial practice explicitly describe dry-laminated battery packaging structures and the use of polyol + isocyanate 2K systems for bonding metal foil to polymer layers while targeting improved formability (deep-drawing resistance). This supports PU as the most relevant baseline chemistry for the proposed R&D [13]. Compared with alternative adhesive families, PU offers a more favorable balance for this application: epoxies often exhibit high stiffness/brittleness; acrylic systems may show lower high-temperature strength; and hot-melt solutions can be limited by creep at elevated temperature. PU provides a broad design window to target both high adhesion and high elongation required for pouch forming [6,7].
Rationale for Joint R&D Project	Achieving both high peel strength (especially Al & PA and PA & PET) and high elongation / toughness is a classic materials trade-off. Higher adhesion frequently correlates with higher crosslink density and modulus, which can reduce elongation and increase crack initiation during forming [6,7]. Optimizing this trade-off requires: (i) molecular-level PU design, (ii) advanced characterization (rheology, DMA, cure kinetics, morphology), and (iii) validation on relevant roll-to-roll lamination and pouch forming processes. This multi-disciplinary loop is difficult for a single SME to execute quickly [4,6,7].

	Build a structure–property–process model linking PU segment design and curing chemistry to peel strength and forming crack resistance [6,7]. Rapidly screen formulations using rheology/DMA/FT-IR-based cure monitoring to shorten iteration cycles and identify failure modes [6,7]. Validate down-selected candidates on pilot lamination and formed-pouch trials, incorporating temperature/humidity sensitivities observed in pouch materials [4]. The Korean SME should ideally be a manufacturer of PU adhesives and aim to expand its product portfolio by leading the applied research on formulation and pouch lamination while the international research institute conducts fundamental research to accelerate the development process. The institute is expected to provide strengths in polymer synthesis and advanced characterization to accelerate structure–property optimization with the following expectations: • Core capabilities: polyurethane/polymer chemistry, adhesion & surface science, rheology/DMA, and high-temperature (120℃) failure analysis. • Preferred contribution: rapid screening, cure/viscoelastic characterization, and mechanism-based formulation guidance.
Key Words	1. Li-ion Battery 2. Polyurethane Adhesive 3. Aluminum Pouch 4. Dry Lamination 5. Forming Depth

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 5

R&D Objectives

1. Define end-user requirements (substrate stack, coating weight, curing window, forming conditions) and establish an agreed test protocol during the pre-planning period [4].
2. Develop 2K PU laminating adhesive formulation(s) suitable for dry lamination of battery pouch film layers (Al & PA and PA & PET), achieving both high peel strength and high elongation/toughness for stable pouch forming [3,4,6,7].
3. Establish a validated process window for pilot lamination (coating weight, drying, nip pressure, line speed, curing/aging) and a structure–property map to guide scale-up [4,6,7].
4. Deliver adhesive batches, laminated film prototypes, and a qualification/commercialization package (specification + QC plan) for customer evaluation.

R&D Target Specifications

	Performance Indicator	Unit	Target	Global Best	Standard/ Certification Criteria
1	forming depth	mm	6.5	6.5(Japan)	Internal pouch forming test: standard forming die; inspect cracks/delamination by optical microscopy; report maximum depth meeting criteria. [3,4]
2	adhesive peel strength between aluminum layer and PA film at 20℃	N/15mm	≥7	7 (Japan)	T-peel per ASTM D1876 or ISO 11339; 23 °C; ≥5 specimens; report average/min/max and failure mode. [9,10]
3	adhesive peel strength between PA film and PET film at 20℃	N/15mm	≥2.0 (prefer PET failure)	2.0 (Prefer PET failure)	T-peel per ASTM D1876 or ISO 11339; 23 °C; ≥5 specimens; report failure mode. [9,10]
4	adhesive peel strength between aluminum layer and PA film at 120℃	N/15mm	≥2.5	2.5 (Japan)	T-peel per ASTM D1876 or ISO 11339 using a temperature chamber at 120 °C (dwell ≥10 min before test). [9,10]
5	adhesive peel strength between PA film and PET film at 120℃	N/15mm	≥2.0 (prefer PET failure)	2.0 (prefer PET failure)	T-peel per ASTM D1876 or ISO 11339 using a temperature chamber at 120 °C. [9,10]

R&D Plan

Work Plan	Period (Month)	Key Tasks & Methodology	Main Deliverables / Milestones
WP0	0–36	Project governance (joint steering, reporting), IP/contracting, EHS, quality management (QC). Establish joint test protocols, data formats, and sample tracking.	Kick-off package; protocol & data management plan; periodic reports; IP strategy; final technical report.
WP1 (Pre-planning)	0–6	Requirement definition with end-users: substrate stack, coating weight, curing window, forming conditions. Benchmark commercial adhesives and failure modes. Finalize test methods (peel at room T/high T, forming depth, electrolyte resistance).	User requirement specification (URS); benchmarking report; agreed test protocol; initial formulation DoE plan.
WP2	4–18	PU chemistry design: select polyol family (polyester/polyether/hybrid), functionality, Mw; tune hard/soft segment ratio; select isocyanate curing agent and catalysts. Characterize viscosity/pot life, cure kinetics (FT-IR NCO decay), T_g and phase separation (DSC/DMA).	1st-gen adhesive candidates (≥5); structure–property map linking formulation variables to peel and elongation.

	WP3	7–24	Adhesion engineering: evaluate Al-oxide interaction, PA/PET surface states (corona/plasma compatibility if applicable). Screen adhesion promoters/coupling agents and additives to improve Al/PA and PA/PET bonding while maintaining toughness.	Down-selected formulations meeting RT peel targets; documented failure modes (cohesive vs interfacial).
	WP4	12–30	Process validation on dry lamination equipment: coating weight, drying temperature/time, nip pressure, line speed, cure/aging. Evaluate forming depth and crack resistance on formed pouches; iterate formulation and process window using DoE.	Laminated film device; stable process window; demonstration of forming depth ≥6.5 mm with no cracking/delamination.
	WP5	18–36	Reliability & qualification screening: high-temperature peel. Prepare qualification data package and draft product specification/QC tests (viscosity, NCO content, pot life, gel time, peel QC).	Qualification report; draft product spec; QC plan; samples for customer qualification.
	WP6	30–36	Commercialization planning: cost model, supply chain, production scale-up roadmap, and customer engagement for adoption.	Go-to-market plan; pilot production plan; partner/customer LOI (if available).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• PU molecular design support: recommend segment architecture and cure chemistry to balance adhesion and elongation. • Formulation guidance: DoE planning, catalyst selection, additives/adhesion promoters screening.
	• Advanced characterization: cure kinetics (FT-IR), viscoelasticity (DMA/rheology), morphology (DSC/SEM as needed). • Perform peel/forming/reliability tests per agreed protocol and compile qualification data. • Failure analysis: identify cracking/delamination mechanisms after forming and propose countermeasures.
Expected R&R of Korean SME Partner	• Define requirements with end-users and manage overall project integration. • Prepare PU adhesive formulations. • Run dry lamination trials and optimize the process window.
	• Secure intellectual property. • Lead commercialization: product spec, cost/production plan, customer qualification, and regulatory/safety documentation.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcomes:** • Validated 2K PU laminating adhesive formulation(s) meeting peel and forming targets for Al&PA and PA&PET interfaces. • Pilot-scale batch recipe and QC specification (viscosity, solids, NCO content, pot life, gel time, cure profile). • Validated lamination/cure process window and failure-mode map for production transfer. • Prototype laminated pouch film rolls and formed pouch samples verified in a relevant environment (pilot lamination + forming). • Qualification data package (RT/HT peel, forming, aging, electrolyte resistance) and draft product specification for customer approval. **Impact:** • Improved yield and reliability in pouch forming by reducing delamination and adhesive cracking defects. • Reduced dependence on imported high-performance laminating adhesives through localization and IP creation. • Strengthened competitiveness of Korean SMEs in the battery materials supply chain via global collaboration and accelerated qualification.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization